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Safety Data Sheet according to Regulation (EC) No. 1907/2006 (REACH)

SECTION 1: Identification of the substance/mixture and of the company/undertaking

* 1.1. Product identifier

Trade name/designation:

RAVENOL VMO SAE 5W-40

Article No.:

1111133

UFI:

44FY-54D2-3CEQ-0HRC

1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the substance/mixture:

Lubricant

1.3. Details of the supplier of the safety data sheet

Supplier (manufacturer/importer/only representative/downstream user/distributor):

Ravensberger Schmierstoffvertrieb GmbH

Produktsicherheit

Jöllenbecker Str. 2

33824 Werther

Germany

Telephone: +49 5203 9719 0

Telefax: +49 5203 9719 40

E-mail: kontakt@ravenol.de

Website: www.ravenol.de

E-mail (competent person): sdb@ravenol.de

* 1.4. Emergency telephone number

24 hr. emergency phone number, 24h: +1 872 5888271 (Contract ID: RAV)

SECTION 2: Hazards identification

* 2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Respiratory or skin sensitisation (Skin Sens. 1)	H317: May cause an allergic skin reaction.	Calculation method.

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms:



GHS07

Exclamation mark

Signal word: Warning



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Hazard components for labelling:

C14-16-18 Alkyl phenol; Molybdenum alkylthiocarbamide complex

Hazard statements for health hazards

H317	May cause an allergic skin reaction.
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Precautionary statements Prevention

P261	Avoid breathing vapours and spray.
P280	Wear protective gloves and eye protection/face protection.

Precautionary statements Response

P302 + P352	IF ON SKIN: Wash with plenty of water and soap.
P333 + P313	If skin irritation or rash occurs: Get medical advice/attention.

Precautionary statements Disposal

P501	Dispose of contents/container to an appropriate recycling or disposal facility.
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2.3. Other hazards

Other adverse effects:

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, annex XIII.

SECTION 3: Composition/information on ingredients

* 3.2. Mixtures

Hazardous ingredients / Hazardous impurities / Stabilisers:

Product identifiers	Substance name Classification according to Regulation (EC) No 1272/2008 [CLP]	Concentration
CAS No.: 64742-54-7 EC No.: 265-157-1 REACH No.: 01-2119484627-25	Distillates (petroleum), hydrotreated heavy paraffinic; Base oil - not specified Asp. Tox. 1 (H304) Danger Acute Toxicity Estimate ATE (oral) 5,000 mg/kg ATE (dermal) 5,000 mg/kg ATE (inhalation, dust/mist) 5.53 mg/L	50 - < 100 weight-%
CAS No.: 1190625-94-5 EC No.: 931-468-2 REACH No.: 01-2119498288-19	C14-16-18 Alkyl phenol Aquatic Chronic 4 (H413), Skin Sens. 1B (H317) Warning Acute Toxicity Estimate ATE (oral) 2,000 mg/kg ATE (dermal) 2,000 mg/kg	0 - < 1.5 weight-%
CAS No.: 36878-20-3 EC No.: 253-249-4 REACH No.: 01-2119488911-28	bis(nonylphenyl)amine Aquatic Chronic 4 (H413) Acute Toxicity Estimate ATE (oral) > 5,000 mg/kg ATE (dermal) > 2,000 mg/kg ATE (inhalation, dust/mist) > 5 mg/L	0 - < 1.5 weight-%
EC No.: 457-320-2 REACH No.: 01-0000019337-66	Molybdenum alkylthiocarbamide complex Aquatic Chronic 3 (H412), Skin Irrit. 2 (H315), Skin Sens. 1B (H317) Warning Acute Toxicity Estimate ATE (oral) > 2,000 mg/kg ATE (dermal) > 2,000 mg/kg	0 - < 0.15 weight-%

Full text of H- and EUH-phrases: see section 16.

SECTION 4: First aid measures

* 4.1. Description of first aid measures

General information:

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Remove victim out of the danger area. Remove contaminated, saturated clothing. If unconscious but breathing normally, place in recovery position and seek medical advice. Do not leave affected person unattended.

Following inhalation:

Provide fresh air. Consult a doctor immediately.



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In case of skin contact:

After contact with skin, wash immediately with plenty of water and soap. Consult a doctor immediately.

After eye contact:

In case of contact with eyes flush immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart and consult an ophthalmologist.

Following ingestion:

Rinse mouth thoroughly with water. Do NOT induce vomiting. Consult a doctor immediately.

Self-protection of the first aider:

Use personal protection equipment. No direct artificial respiration to be given by first aider.

* **4.2. Most important symptoms and effects, both acute and delayed**

May produce an allergic reaction.

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically. Observe risk of aspiration if vomiting occurs.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

Co-ordinate fire-fighting measures to the fire surroundings.

Carbon dioxide (CO₂)

Extinguishing powder

alcohol resistant foam

Use water spray jet to protect personnel and to cool endangered containers.

Unsuitable extinguishing media:

Full water jet

5.2. Special hazards arising from the substance or mixture

During heating or in case of fire, toxic gases is possible.

The formation of combustible vapours is possible at temperatures above: Flash point

When hot, product develops flammable vapours.

Hazardous combustion products:

Carbon monoxide, Carbon dioxide (CO₂), Nitrogen oxides (NO_x), Gases/vapours, toxic

During heating or in case of fire, toxic gases is possible.

5.3. Advice for firefighters

In case of fire: Wear self-contained breathing apparatus. Protective clothing.

5.4. Additional information

Do not inhale explosion and combustion gases. Move undamaged containers from immediate hazard area if it can be done safely. Collect contaminated fire extinguishing water separately. Do not allow entering drains or surface water.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Personal precautions:

Use personal protection equipment. Special danger of slipping by leaking/spilling product.

Protective equipment:

Personal protection equipment: see section 8

Emergency procedures:

Eliminate all ignition sources if safe to do so. Remove persons to safety. Provide adequate ventilation.

6.1.2. For emergency responders

Personal protection equipment:

Use personal protection equipment.

6.2. Environmental precautions

Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains. Prevent spread over a wide area (e.g. by containment or oil barriers). In case of gas escape or of entry into waterways, soil or drains, inform the responsible authorities.



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6.3. Methods and material for containment and cleaning up

For containment:

Suitable material for taking up: Sand, Kieselguhr, Universal binder, Chemical binding agents, containing acids
Prevent spread over a wide area (e.g. by containment or oil barriers).

For cleaning up:

Remove from the water surface (e.g. skimming, sucking). Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

Other information:

Treat the recovered material as prescribed in the section on waste disposal.

6.4. Reference to other sections

Safe handling: see section 7
Disposal: see section 13
Personal protection equipment: see section 8

6.5. Additional information

Clear spills immediately. Use appropriate container to avoid environmental contamination.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Protective measures

Advices on safe handling:

Wear personal protection equipment (refer to section 8).
When using do not eat, drink, smoke, sniff. Wash hands before breaks and after work. Do not put any product-impregnated cleaning rags into your trouser pockets. Clear spills immediately. Use appropriate container to avoid environmental contamination.

Fire prevent measures:

No special fire protection measures are necessary.

Environmental precautions:

Shafts and sewers must be protected from entry of the product.

Advices on general occupational hygiene

Minimum standard for preventive measures while handling with working materials are specified in the TRGS 500.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions:

Keep container tightly closed in a cool, well-ventilated place.

Requirements for storage rooms and vessels:

Suitable container/equipment material: Floors should be impervious, resistant to liquids and easy to clean. Shafts and sewers must be protected from entry of the product.
Keep/Store only in original container.

Hints on storage assembly:

not required

Storage class (TRGS 510, Germany): 10 – Combustible liquids that cannot be assigned to any of the above storage classes

Further information on storage conditions:

Store in a cool dry place. Keep away from heat.

7.3. Specific end use(s)

Recommendation:

Observe technical data sheet.

SECTION 8: Exposure controls/personal protection

* 8.1. Control parameters

8.1.1. Occupational exposure limit values

No data available

8.1.2. Biological limit values

No data available



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8.1.3. DNEL-/PNEC-values

Substance name	DNEL value	① DNEL type ② Exposure route
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	1.17 mg/m ³	① DNEL worker ② Long-term - inhalation, systemic effects
bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4	5 mg/kg bw/day	① DNEL worker ② Long-term - dermal, systemic effects
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	3.52 mg/m ³	① DNEL worker ② Long-term - inhalation, systemic effects
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	2.24 mg/kg	① DNEL worker ② Long-term - dermal, systemic effects
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	0.112 mg/cm ²	① DNEL worker ② Long-term - dermal, local effects

Substance name	PNEC Value	① PNEC type
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	100 µg/L	① PNEC aquatic, freshwater
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	10 µg/L	① PNEC aquatic, marine water
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	100 mg/L	① PNEC sewage treatment plant
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	4,266.16 mg/kg bw/day	① PNEC sediment, freshwater
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	426.62 mg/kg bw/day	① PNEC sediment, marine water
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	852.58 mg/kg bw/day	① PNEC soil
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	3.3 mg/kg bw/day	① PNEC secondary poisoning
C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2	1 mg/L	① PNEC aquatic, intermittent release
bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4	412 µg/L	① PNEC aquatic, freshwater
bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4	41.2 µg/L	① PNEC aquatic, marine water
bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4	1 mg/L	① PNEC aquatic, intermittent release
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	0.081 mg/L	① PNEC aquatic, freshwater
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	0.008 mg/L	① PNEC aquatic, marine water
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	10 mg/L	① PNEC sewage treatment plant
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	195 mg/kg	① PNEC sediment, freshwater



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Substance name	PNEC Value	① PNEC type
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	19.5 mg/kg	① PNEC sediment, marine water
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	0.872 mg/kg	① PNEC soil
Molybdenum alkylthiocarbamide complex EC No.: 457-320-2	20 mg/kg	① PNEC secondary poisoning

8.2. Exposure controls

8.2.1. Appropriate engineering controls

See section 7. No additional measures necessary.

8.2.2. Personal protection equipment



Eye/face protection:

During transfer: Eye glasses with side protection

Wear eye/face protection. EN 166

Skin protection:

Hand protection

Suitable material: NBR (Nitrile rubber), PVC (polyvinyl chloride), CR (polychloroprene, chloroprene rubber)

Thickness of the glove material: $\geq 0,4$ mm

Breakthrough time: 480 min

Breakthrough times and swelling properties of the material must be taken into consideration.

The quality of the protective gloves resistant to chemicals must be chosen as a function of the specific working place concentration and quantity of hazardous substances.

For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

Tested protective gloves must be worn: EN ISO 374

Suitable protective clothing: Protective clothing

Respiratory protection:

Usually no personal respirative protection necessary.

8.2.3. Environmental exposure controls

See section 7. No additional measures necessary.

SECTION 9: Physical and chemical properties

* 9.1. Information on basic physical and chemical properties

Appearance

Physical state: Liquid

Form: Liquid

Colour: tawny

Odour: characteristic

flammability: Yes

Safety relevant basis data

Parameter	Value	at °C	① Method ② Remark
pH	<i>not applicable</i>		
Melting point	<i>No data available</i>		
Freezing point	<i>No data available</i>		
Initial boiling point and boiling range	<i>No data available</i>		
Decomposition temperature	<i>not applicable</i>		
Flash point	240 °C		
Evaporation rate	<i>No data available</i>		
Auto-ignition temperature	<i>No data available</i>		
Upper/lower flammability or explosive limits	<i>No data available</i>		
Vapour pressure	<i>No data available</i>		



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Parameter	Value	at °C	① Method ② Remark
Vapour density	No data available		
Density	850 kg/m³	15 °C	
Relative density	not applicable		
Bulk density	not applicable		
Water solubility	practically insoluble		
Partition coefficient: n-octanol/water	not applicable		
Dynamic viscosity	No data available		
Kinematic viscosity	77 mm²/s	40 °C	

9.2. Other information

Not applicable

SECTION 10: Stability and reactivity

10.1. Reactivity

No known hazardous reactions.

10.2. Chemical stability

The mixture is chemically stable under recommended conditions of storage, use and temperature.

10.3. Possibility of hazardous reactions

No hazardous reaction when handled and stored according to provisions.

10.4. Conditions to avoid

To avoid thermal decomposition do not overheat.

10.5. Incompatible materials

Materials to avoid: Acid, Oxidizing agent, Reducing agent

10.6. Hazardous decomposition products

Hazardous combustion products: Carbon monoxide, Carbon dioxide (CO₂), Nitrogen oxides (NO_x),
 During heating or in case of fire, toxic gases is possible.

Further information

No information available.

SECTION 11: Toxicological information

* 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Toxicological information

Acute Toxicity Estimate for Mixtures	
ATE (oral): 148,149.6 mg/kg	
ATE (dermal): 74,074.4 mg/kg	
ATE (inhalation, dust/mist): 370.37 mg/L	
Distillates (petroleum), hydrotreated heavy paraffinic; Base oil - not specified	CAS No.: 64742-54-7 EC No.: 265-157-1
LD₅₀ oral: 5,000 mg/kg (Rat) OECD 401	
LD₅₀ dermal: 5,000 mg/kg (Rabbit) OECD 402	
LC₅₀ Acute inhalation toxicity (dust/mist): 5.53 mg/L 4 h (Rat) OECD 403	
C14-16-18 Alkyl phenol	CAS No.: 1190625-94-5 EC No.: 931-468-2
LD₅₀ oral: 2,000 mg/kg	
LD₅₀ dermal: 2,000 mg/kg	
bis(nonylphenyl)amine	CAS No.: 36878-20-3 EC No.: 253-249-4
LD₅₀ oral: >5,000 mg/kg (Rat)	
LD₅₀ dermal: >2,000 mg/kg (Rabbit)	
LC₅₀ Acute inhalation toxicity (dust/mist): >5 mg/L	
Molybdenum alkylthiocarbamide complex	EC No.: 457-320-2
LD₅₀ oral: >2,000 mg/kg (Rat) OECD Guideline 425	
LD₅₀ dermal: >2,000 mg/kg (Rat) OECD Guideline 402	



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Acute oral toxicity:

Based on available data, the classification criteria are not met.

Acute dermal toxicity:

Based on available data, the classification criteria are not met.

Acute inhalation toxicity:

Based on available data, the classification criteria are not met.

Skin corrosion/irritation:

Based on available data, the classification criteria are not met.

Serious eye damage/irritation:

Based on available data, the classification criteria are not met.

Respiratory or skin sensitisation:

May cause an allergic skin reaction.

Germ cell mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity:

Based on available data, the classification criteria are not met.

STOT-single exposure:

Based on available data, the classification criteria are not met.

STOT-repeated exposure:

Based on available data, the classification criteria are not met.

Aspiration hazard:

Observe risk of aspiration if vomiting occurs.

For viscosity data, see section 9.

Additional information:

Frequently or prolonged contact with skin may cause dermal irritation.

11.2. Information on other hazards

Endocrine disrupting properties:

This product does not contain a substance that has endocrine disrupting properties with respect to humans as no components meets the criteria.

Other information:

No data available.

SECTION 12: Ecological information

* **12.1. Toxicity**

Distillates (petroleum), hydrotreated heavy paraffinic; Base oil - not specified CAS No.: 64742-54-7
EC No.: 265-157-1

LC₅₀: 100 mg/L 4 d (fish)

LC₅₀: 10,000 mg/L 4 d (crustaceans)

EC₅₀: 10,000 mg/L 2 d (crustaceans)

NOEC: 100 mg/L 4 d (fish)

NOEC: 100 mg/L 3 d (Algae/water plant)

NOEC: ≥100 mg/L 3 d (Algae/water plant, Algen)

C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2

EC₅₀: 100 mg/L 2 d (crustaceans)

EC₅₀: 100 mg/L 2 d (Algae/water plant)

bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4

LC₅₀: >100 mg/L 4 d (fish)

EC₅₀: >100 mg/L 2 d (crustaceans)

EC₅₀: 600 mg/L 3 d (Algae/water plant)



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Molybdenum alkylthiocarbamide complex EC No.: 457-320-2

LC₅₀: 670 mg/L 4 d (fish)

EC₅₀: 50 mg/L 2 d (crustaceans)

EC₅₀: 9.62 mg/L 3 d (Algae/water plant)

NOEC: 100 mg/L 21 d (crustaceans)

NOEC: 4.05 mg/L 3 d (Algae/water plant)

Assessment/classification:

Based on available data, the classification criteria are not met.

Additional ecotoxicological information:

Do not allow uncontrolled discharge of product into the environment.

* **12.2. Persistence and degradability**

bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4

Biodegradation: —

Biodegradation:

Not readily biodegradable (according to OECD criteria)

* **12.3. Bioaccumulative potential**

bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4

Log K_{ow}: 7.6

Bioconcentration factor (BCF): 1,584.89

Partition coefficient: n-octanol/water:

not applicable

Accumulation / Evaluation:

The product has not been tested.

12.4. Mobility in soil

The product has not been tested.

* **12.5. Results of PBT and vPvB assessment**

Distillates (petroleum), hydrotreated heavy paraffinic; Base oil - not specified CAS No.: 64742-54-7
 EC No.: 265-157-1

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

C14-16-18 Alkyl phenol CAS No.: 1190625-94-5 EC No.: 931-468-2

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

bis(nonylphenyl)amine CAS No.: 36878-20-3 EC No.: 253-249-4

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

Molybdenum alkylthiocarbamide complex EC No.: 457-320-2

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, annex XIII.

12.6. Endocrine disrupting properties

This product does not contain a substance that has endocrine disrupting properties with respect to non-target organisms as no components meets the criteria.

12.7. Other adverse effects

No data available.

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Dispose of waste according to applicable legislation.

Waste treatment options

Appropriate disposal / Product:

Dispose of waste according to applicable legislation.

Appropriate disposal / Package:

Non-contaminated packages may be recycled.

Other disposal recommendations:

Consult the appropriate local waste disposal expert about waste disposal.



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13.2. Additional information

The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

SECTION 14: Transport information

Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.1. UN number or ID number			
No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.
14.2. UN proper shipping name			
No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.
14.3. Transport hazard class(es)			
not relevant	not relevant	not relevant	not relevant
14.4. Packing group			
not relevant	not relevant	not relevant	not relevant
14.5. Environmental hazards			
not relevant	not relevant	not relevant	not relevant
14.6. Special precautions for user			
not relevant	not relevant	not relevant	not relevant

14.7. Maritime transport in bulk according to IMO instruments

Not applicable

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU legislation

Other regulations (EU):

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances [Seveso-III-Directive]: This product is not assigned to a hazard category.

Safety data sheet available on request.

Directive 2004/42/EC on the limitation of emissions of volatile organic compounds:

Volatile organic compound (VOC) content: 0 weight-%

15.1.2. National regulations

[DE] National regulations

Restrictions of occupation

Observe restrictions to employment for juveniles according to the 'juvenile work protection guideline' (94/33/EC). Observe employment restrictions under the Maternity Protection Directive (92/85/EEC) for expectant or nursing mothers.

Störfallverordnung (12. BImSchV)

for substances contained in the product:

This product is not assigned to a hazard category.

Technische Anleitung zur Reinhaltung der Luft (TA-Luft)

Remark:

To follow: 5.2.5

Water hazard class

WGK:

1 - slightly hazardous to water

Source:

Self-classification (mixture; calculation rule).

Identification number 436

Technische Regeln für Gefahrstoffe

TRGS 510

TRGS 500



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Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Berufsgenossenschaftliche Informationen (DGUV-Informationen) 868
Berufsgenossenschaftliche Regeln (DGUV-Regeln) 189, 190, 192, 195

Other regulations, restrictions and prohibition regulations

Altöl-Verordnung (AltöIV)



[DK] National regulations

Other regulations, restrictions and prohibition regulations

Dänemark: Bekendtgørelse af lov om arbejdsmiljø: Beskæftigelsesministeriets lovbekendtgørelse nr. 1072 af 7. september 2010
Lister over stoffer og processer, der anses for at være kræftfremkaldende



[FR] National regulations

Other regulations, restrictions and prohibition regulations

Frankreich: Tableaux de maladies professionnelles
Nomenclature des installations classées pour la protection de l'environnement
Articles L. 4523-1 à L. 4523-17, L. 4611-1 à L. 4614-16, R. 4523-1 à R. 4523-17 et R. 4612-1 à R. 4615-21 du Code du travail



[NL] National regulations

Other regulations, restrictions and prohibition regulations

Niederlande: Lijst van kankerverwekkende, mutagene en voor de voortplanting giftige stoffen (SZW)
Algemeene beoordelingsmethodiek Water (ABM)
Nederlandse emissierichtlijn (NeR)
NIET-Limitatieve lijst aan voor de voortplanting giftige stoffen - Borstvoeding
NIET-Limitatieve lijst aan voor de voortplanting giftige stoffen - Vruchtbaarheid
NIET-Limitatieve lijst aan voor de voortplanting giftige stoffen - Ontwikkeling
SZW-lijst van kankerverwekkende stoffen
SZW-lijst van mutagene stoffen
Wet van 18 maart 1999, houdende bepalingen ter verbetering van de arbeidsomstandigheden (Arbeidsomstandighedenwet)
Wet op de ondernemingsraden 1971



[CH] National regulations

Other regulations, restrictions and prohibition regulations

Mengenschwelle (Schweiz - StFV)
Gefahrencode
Brandverhütung, BVD (Schweiz)



[SK] National regulations

Other regulations, restrictions and prohibition regulations

Zákon č. 67/2010 Z.z., o podmienkach uvedenia chemických látok a chemických zmesí na trh a o zmene a doplnení niektorých zákonov (chemický zákon).
Zákon č. 124/2006 Z. z. o bezpečnosti a ochrane zdravia pri práci a o zmene a doplnení niektorých zákonov.
Zákon NR SR č. 355/2007 Z.z., o ochrane, podpore a rozvoji verejného zdravia a o zmene a doplnení niektorých zákonov, v znení neskorších predpisov.
Nariadenie vlády SR 471/2011 Z.z., ktorým sa mení nariadenie vlády Slovenskej republiky č. 355/2006 Z. z. o ochrane zamestnancov pred rizikami súvisiacimi s expozíciou chemickým faktorom pri práci, Príloha č.1.
Zákon č. 79/2015 Z.z. o odpadoch v znení neskorších predpisov.
Vyhláška MV SR č. 96/2004 Z.z., ktorou sa ustanovujú zásady protipožiarnej bezpečnosti pri manipulácii a skladovaní horľavých kvapalín, ťažkých vykurovacích olejov a rastlinných a živočíšnych tukov a olejov.
Zákon NR SR č. 137/2010 Z.z. o ovzduší v znení neskorších predpisov.
Zákon č. 319/2013 Z.z. o pôsobnosti orgánov štátnej správy pre sprístupňovanie biocídnych výrobkov na trh a ich používanie a o zmene a doplnení niektorých zákonov (biocídny zákon).

15.2. Chemical Safety Assessment

Chemical safety assessments for substances in this mixture were not carried out.

15.3. Additional information

No data available.



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SECTION 16: Other information

* 16.1. Indication of changes

1.1.	Product identifier
1.4.	Emergency telephone number
2.1.	Classification of the substance or mixture
3.2.	Mixtures
4.1.	Description of first aid measures
4.2.	Most important symptoms and effects, both acute and delayed
8.1.	Control parameters
9.1.	Information on basic physical and chemical properties
11.1.	Information on hazard classes as defined in Regulation (EC) No 1272/2008
12.1.	Toxicity
12.2.	Persistence and degradability
12.3.	Bioaccumulative potential
12.5.	Results of PBT and vPvB assessment
14.3.	Transport hazard class(es)
15.1.	Safety, health and environmental regulations/legislation specific for the substance or mixture
16.1.	Indication of changes
16.2.	Abbreviations and acronyms
16.4.	Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]
16.5.	List of relevant hazard statements and/or precautionary statements from sections 2 to 15

* 16.2. Abbreviations and acronyms

ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute Toxicity Estimate
BCF	Bioconcentration Factor
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging
DNEL	derived no-effect level
EC ₅₀	Effective Concentration 50%
EG	European Community
EU	European Union
ICAO	International Civil Aviation Organization
IMDG	International Maritime Dangerous Goods
IMO	International Maritime Organization
KG	body weight
LC ₅₀	Lethal (fatal) Concentration 50%
LD ₅₀	Lethal (fatal) Dose 50%
NFPA	National Fire Protection Association
NOEC	No Observed Effect Concentration
OECD	Organisation for Economic Cooperation and Development
PBT	persistent and bioaccumulative and toxic
PNEC	Predicted No Effect Concentration
REACH	Registration, Evaluation and Authorization of Chemicals
RID	Dangerous goods regulations for transport by rail
Tox.	Toxicity
TRGS	Technische Regeln für Gefahrstoffe
UFI	Unique Formula Identifier
UN	United Nations
VOC	Volatile organic compounds
See overview table at www.euphrac.eu	
For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemical safety assessment, chapter R.20 (Table of terms and abbreviations).	

16.3. Key literature references and sources for data

EC 1907/2006 - REACH Regulation
 1272/2008 EC - Regulation on classification, labeling and packaging of substances and mixtures, and amending Directives 67/548/EEC and 1999/45/EC and Regulation (EC) No 1907/2006
 Regulation (EC) No 1907/2006 (REACH), Annex II



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European Chemicals Agency (ECHA), C & L classification and labeling inventory
 European Chemicals Agency (ECHA), ECHA CHEM Registered substances
 OECD The Global Portal to Information on Chemical Substances (ChemPortal)
 Institute for Occupational Safety and Health of the German Social Accident Insurance (IFA): GESTIS
 substance database and International limit values for chemical substances
 Federal Environment Agency, Section IV 2.4: Documentation and Information Centre substances
 hazardous to water Rigoletto (catalog substances hazardous to water)

* **16.4. Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]**

Hazard classes and hazard categories	Hazard statements	Classification procedure
Respiratory or skin sensitisation (Skin Sens. 1)	H317: May cause an allergic skin reaction.	Calculation method.

* **16.5. List of relevant hazard statements and/or precautionary statements from sections 2 to 15**

Hazard statements	
H304	May be fatal if swallowed and enters airways.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H412	Harmful to aquatic life with long lasting effects.
H413	May cause long lasting harmful effects to aquatic life.

16.6. Training advice

No data available

16.7. Additional information

The above information describes exclusively the safety requirements of the product and is based on our present-day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material.

* Data changed compared with the previous version.